



ADNAN KHAN

Operations Research | Stochastic Optimization | Supply Chain Analytics

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📍 Paris, France 🏠 GitHub (click me) 📅 October 2026

🎓 EDUCATION

- Operations Research & Risk Analytics (M2), CentraleSupélec** 09/2025 – 10/2026 | Paris, France
• Grade: 14/20 (Current Average) | Institutional Scholarship Holder
- Industrial Engineering (B.Eng), University of Dhaka** 04/2020 – 12/2024 | Dhaka, Bangladesh
• CGPA: 3.50/4.00 | 87th percentile in the department.

🧠 TECHNICAL SKILLS

- Solvers and Modelling** — Gurobi, Pyomo, Excel Solver
Programming and Data — C/C++, Python (NumPy, Pandas, SciPy, Matplotlib), SQL, Power BI
Optimization — MILP, Duality, Sensitivity Analysis, SAA, Benders Decomposition, Vehicle Routing, Network Design

📁 EXPERIENCE

- Institut du Cerveau (Paris Brain Institute), Intern, Modelling and Data Analysis** 06/2026 – Present | Paris, France
Gaussian HMMs and analysis pipelines in Python; maximum likelihood estimation and Viterbi decoding on experimental time series to extract latent states, transition probabilities and dwell times.
- Akij Food and Beverage Limited (AFBL), Industrial Engineering Intern** 05/2024 – 07/2024 | Bangladesh
Time and motion studies across 4 layouts to locate throughput bottlenecks; redesigned production layout using LEAN.
- Kabir Steel Re-Rolling Mills (KSRM), Production Intern** 06/2023 – 08/2023 | Chattogram, Bangladesh
Cut production defects 11% over a 27-day tracked cycle; Pareto and 5-Why root cause analysis to isolate dominant failures.

📁 PROJECTS

- Optimization of Maintenance Planning for Offshore Wind Turbines** 11/2025 – 12/2025
Python (Pyomo, HiGHS) | MILP, Maintenance Scheduling, Reliability
• MILP scheduling preventive and corrective maintenance over a 30 week horizon under crew capacity, one farm per crew per week, and a one week travel lockout on transfers; solved to optimality on small, medium and large instances.
• Optimum was zero intervention on all three; traced it to a break even of 37.5 weeks against a 30 week horizon (75€ repair plus transport versus 2€/week downtime), the given cost structure made the model ill posed for preventive action.
- Capacity Planning and Machine Replacement under Uncertainty** 03/2026 – 04/2026
Stochastic Programming, Benders Decomposition, CVaR, Markov Decision Processes
• Capacity planning: two-stage stochastic LP with recourse, solved as an SAA extensive form and by Benders L-shaped decomposition with the two cross-checked against each other; extended to a risk-averse CVaR objective.
• Machine replacement: infinite-horizon MDP over ten deterioration states, value iteration converging in 533 sweeps to a control limit policy that keeps in states 1 to 7 and replaces from state 8; dual LP over discounted occupancy measures priced reliability at about 0.65 k€ per 1% output increase.
- Food Bank Inventory Management under Stochastic Demand** 01/2026 – 02/2026
Two-Stage SAA, Network Optimization, Equity Constraints, Python
• Extensive-form SAA over 260 demand weeks setting weekly orders for 2 warehouses and the daily 7x2x6 shipment tensor; sparse LP solved with HiGHS. 93.8% fill rate at a binding 10% cap on inter-district satisfaction spread.
• Quantile forecast policy tuned by leave-one-year-out CV ranked on fairness violations before cost: 99.2% mean fill rate, 1.7% equity gap, zero violations out of sample.

📄 THESIS

- Integrated Vehicle Routing, Order Assignment and Optimization for Resilient Omnichannel Distribution Networks,** CentraleSupélec | Supervisor: Prof. Jakob Puchinger | MILP, Vehicle Routing, Python (Pyomo, Gurobi)
• Two-stage stochastic MILP pre-committing backup fulfillment nodes across 3 channels, with routing, 3PL and unserved recourse under 7 disruption scenarios; 14,334 variables, solved in Gurobi to a 1.9% gap.
• Backup layer cut expected cost 32.3% and unserved demand from 5 to 2 units; stochastic planning cut expected cost 61% against the deterministic plan.

🏆 COMPETITIONS

- TOP 10 teams in the BIHRM Supply Chain Contest, achieved 75% merit-based scholarship in SC professional program.
• Regionally selected and competed at the national level in the Bangladesh Math Olympiad (2016).

🌐 LANGUAGES

English - C1 Level (British Council) French - B1 Level (FLE) Bengali - Native Level